32 Decoding the Palimpsest: An Open Computational Imaging Protocol for Weathered Petroglyphs at Shokur, Oman

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Documenting highly weathered petroglyphsin excavated contexts poses a significant diagnostic challenge. Conventional photography is often insufficient, while high-end 3D scanning equipment remainsinaccessible for many cultural heritage professionals. This contribution presents a reliable computational imaging protocol developed for the diagnostic reading of poorly legible engraved surfaces. The potential of this approach isillustrated through the analysis of a fragmented limestone slab (c. 1.00 × 0.90 m) recovered beside a Bronze Age monumental tower at Shokur (Ad Dhahirah Governorate, Oman).

Establishing a verifiable diagnostic procedure required an end-to-end Free/Libre and Open Source (FLOS) pipeline designed to yield FAIR-compliant datasets. The methodology integrates node-based photogrammetry (Meshroom/AliceVision) to translate standard digital photography into metrically accurate 3D models, Blender-driven Virtual Reflectance Transformation Imaging (RTI) for interactive micro-relief exploration, and Multi-Scale Integral Invariant (MSII) curvature analysis via the GigaMesh Software Framework to separate faint anthropogenic pecked tracesfrom post-depositional damage.

Crucially, this protocol avoidsreliance on proprietary, "black-box" enhancement algorithms. To ensure pipeline quality certification, the interpretative processrelies on the geometric convergence of independent datasets. Candidate traces were retained and manually vectorised (via Inkscape) only when corroborated across photographic, RTI, and curvature-based outputs.

Applied to the Shokurslab, this workflow successfully identified schematic quadrupeds and a possible mounted figure, allowing for the reconstruction of a complex petroglyph palimpsest. More broadly, this case study demonstrates how standard field equipment, paired with a rigorously structured FLOS pipeline, provides a highly adaptable and scientifically robust solution for remote sensing across diverse cultural heritage contexts.